Coding & Solution

Date	15-07-2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution. The Rising Waters – Flood Prediction System Using Machine Learning project demonstrates a functional web-based application that predicts flood occurrence using machine learning techniques. The solution includes data preprocessing, model training, prediction, visualization, and a user-friendly dashboard.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/yourusername/Rising-Waters-Flood-Prediction (Replace with your GitHub repository link)
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Chart.js
Key Features Implemented	• Flood prediction using Machine Learning • Data preprocessing and feature selection • Multiple ML models (Decision Tree, Random Forest, KNN, XGBoost) • Automatic best model selection • Interactive Flask web dashboard • Flood risk percentage calculation • Graphical visualization using Chart.js
Pending / Incomplete Features	• User authentication and login system • Real-time weather API integration • SMS/Email flood alert notifications • Cloud deployment and live monitoring
Setup / Run Instructions	1. Install required packages: pip install -r requirements.txt 2. Train the model: python train_model.py 3. Run the Flask application: python app.py 4. Open the browser and visit: http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (After uploading to GitHub)

Additional Notes / Comments

- The application follows a modular architecture by separating the presentation layer, application logic, and machine learning model.
- The project uses multiple machine learning algorithms and automatically selects the best-performing model based on evaluation metrics.
- The web interface provides an intuitive dashboard for entering flood-related parameters and displaying prediction results with visual charts.
- The code is structured for maintainability and can be extended with features such as real-time weather data integration, cloud deployment, and notification services.
- The project is suitable for academic demonstration and can be enhanced for practical disaster management applications.